HRRP

Housing Recovery and Reconstruction Platform

Housing Typologies: Earthquake Affected Districts

September 2018

About HRRP

The Housing Recovery and Reconstruction Platform (HRRP) was established in December 2015 to take over supporting coordination of the post-earthquake housing reconstruction from the Nepal Shelter Cluster, as it returned to the pre-earthquake format as a standing cluster. The platform provides coordination support services for the National Reconstruction Authority (NRA), Building and Grant Management and Local Infrastructure (GMALI) Central Level Programme Implementation Units (CLPIUs), other relevant government authorities, and Partner Organisations (POs). Phase 3 of the HRRP was approved by the Government of Nepal (GoN) at the beginning of March 2017 and will run until the end of February 2019. HRRP3 is primarily funded by DFID Nepal and CRS Nepal. Other financial contributors and implementing partners include Oxfam, Caritas Nepal, Plan International, National Society for Earthquake Technology-Nepal (NSET), and Habitat for Humanity.

The HRRP has 12 District Coordination Teams (DCTs) primarily focused on the 14 districts most affected by the 2015 Gorkha earthquake (1 team covers the three districts in the Kathmandu Valley) and providing support to the 18 moderately affected districts where feasible. The DCTs are made up of a Coordinator, a Technical Coordinator, and an Information Management Officer. The DCTs are supported by a District Management Team (DMT) made up of a Coordinator, Technical Coordinator, and Information Manager. The DMT provides day to day guidance and support to the DCTs as well as targeted capacity building and has a roving presence across all districts. The national team includes general coordination, technical coordination, and information management expertise and supports the link between national and district level.

Areas of Focus

The HRRP has four main areas of focus:

- Monitoring and documenting the housing reconstruction process
- Improving coverage and quality of socio-technical assistance
- Addressing gaps and duplications
- Advocacy and Communications

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Introduction

This document provides examples of the wide range of housing types being constructed across the 32 districts affected by the April 2015 Gorkha earthquake. The examples presented in this document have been collected by HRRP District Coordination Teams (DCTs), either directly during field visits or sourced from Government of Nepal (GoN) and Partner Organisation (PO) staff.

There are 13 main typologies that the examples collected from the districts can be grouped under:

1. Stone and Cement Mortar Masonry (SCM)
2. Stone and Mud Mortar Masonry (SMM)
3. Brick and Cement Mortar Masonry (BCM)
4. Brick and Mud Mortar Masonry (BMM)
5. Reinforced Cement Concrete Framed Building (RCC)
6. Hybrid Structures
7. Timber Frame Structures
8. Hollow Concrete Block and Cement Mortar Masonry (HCB)
9. Dry Stone Masonry (DSM)
10. Adobe Structures
11. Bamboo Structures
12. Compressed Stabilised Earth Block (CSEB) Masonry
13. Light Steel Frame Structures

This document provides examples for each of these typologies and all photos of housing typologies collected by HRRP can be accessed through the HRRP Flickr account: https://www.flickr.com/photos/hrrp. HRRP would like to extend our gratitude to those that have already shared photos – they are an invaluable resource – and we would encourage all our partners to continue to share photos when possible.

The beginning of each of the sections in the document includes a map showing, by district, the percentage of buildings of that typology damaged by the earthquake. This data has been sourced from the Central Bureau of Statistics (CBS) damage assessment, which was carried out to determine eligibility for the government's housing reconstruction or retrofit grants.

Each section also includes a table of 'pros' and 'cons'. These are not intended to provide a value judgement of any particular typology in relation to others, but simply to provide an indication of the positive and negative aspects of the different typologies that households may face during their reconstruction.

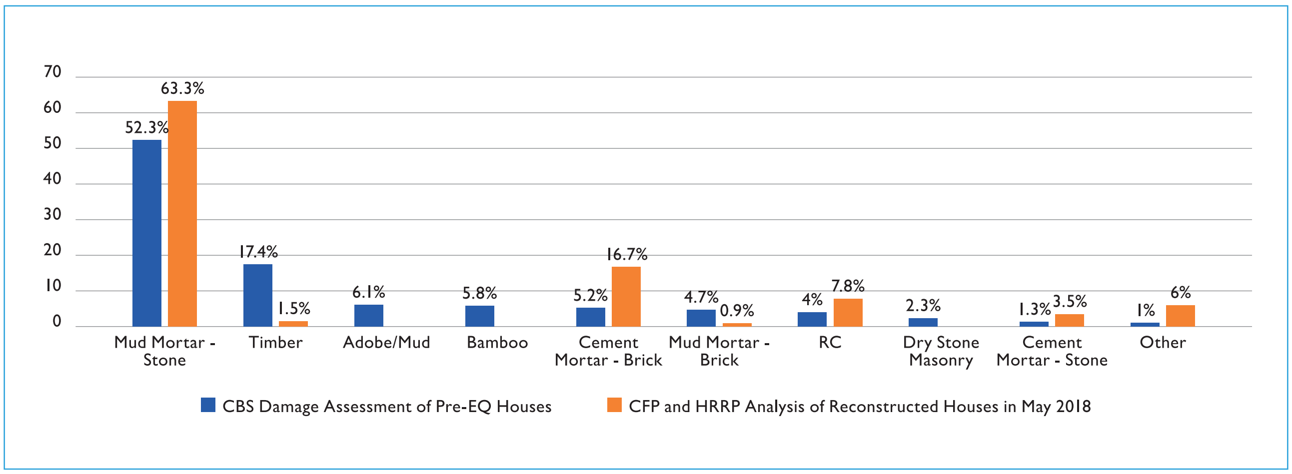
The photo captions are structured as follows:

- **Location:** District, Rural / Urban Municipality, Ward No.
- **Tranche Status:** tranches received by the house owner to date (if information known)
- **Other:** any other information available about the house
- **Photo Credit:** provided if photo not taken by HRRP team

Background

Typologies Pre and Post-Earthquake

The graph below presents a comparison of the prevalence of typologies pre and post earthquake. The pre-earthquake data is from the Central Bureau of Statistics (CBS) damage assessment which determined eligibility for the GoN reconstruction or retrofit grants. The post-earthquake data is from an analysis carried out by HRRP and the Inter-Agency Common Feedback Project (CFP) in May 2018. For this, the HRRP Technical Coordination team analysed more than 500 photos taken by CFP enumerators during the survey to assess typology, size, and compliance. Obviously, this cannot be a direct comparison with the CBS data due to the small sample size but is presented as an indication only. In the graph, 'other' refers to Hollow Concrete Block (HCB) and hybrid structures.



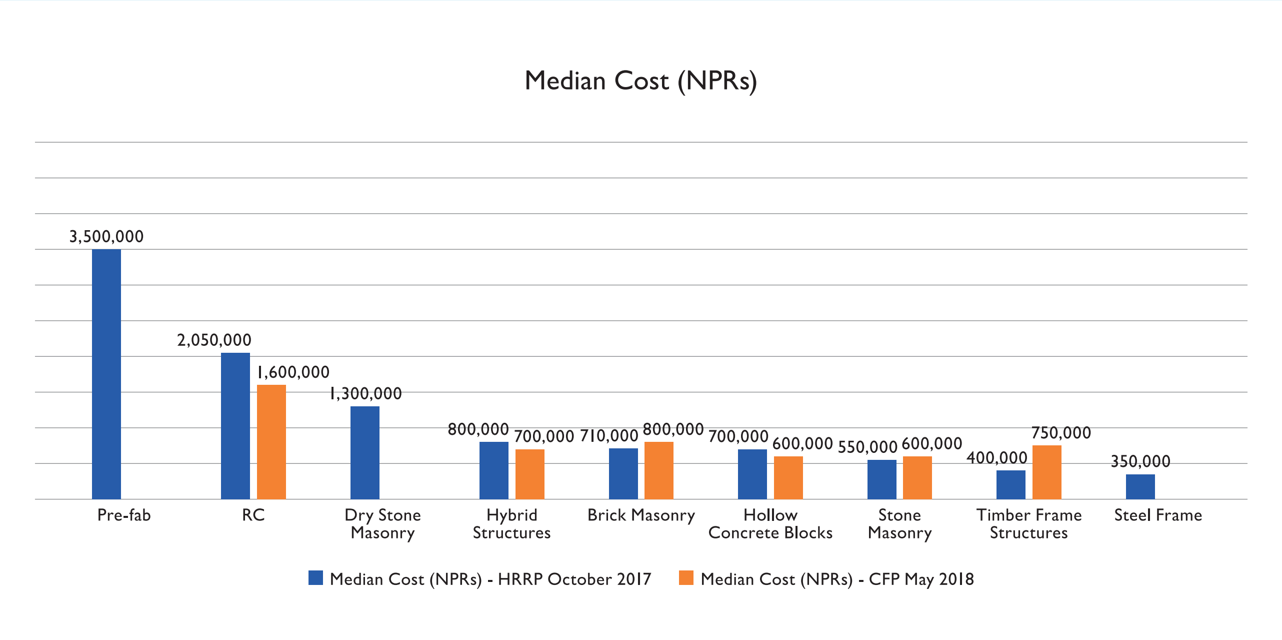
Cost of Construction

In October 2017, the HRRP District Coordination Teams (DCTs) spoke by telephone with 570 households who had previously taken part in Inter-Agency Common Feedback Project surveys to collect data on construction costs. They asked three questions:

1. How much did your house cost / do you estimate it will cost?
2. What building materials did you use / are you planning to use, and have you salvaged materials?
3. Do you plan to / have you already taken a loan and if so for how much, from what type of financial institution, and at what interest rate?

The information provided by the respondents provided indicative data only as it was not a statistically representative sample size. However, it offers useful insight into the financial aspects of reconstruction around which there is a limited amount of existing data. The overall median cost of construction was found to be 675,000 NPRs (Nepalese Rupees). In May 2018, the Inter-Agency Common Feedback Project (CFP) asked participants in community feedback surveys 'how much money do you anticipate needing to finish your overall housing reconstruction?' and the median from this was 700,000 NPRs.

The graph below presents a comparison of the median costs by typology, from the HRRP data collection in October 2017 and from the CFP survey in May 2018.



Government of Nepal Housing Reconstruction Programme

The Government of Nepal (GoN) began planning the owner driven housing reconstruction programme in June 2015 after the launch of the Post Disaster Needs Assessment (PDNA). The programme includes a housing reconstruction grant of 300,000 NPRs, provided by the GoN in three tranches linked to compliant construction; 50,000 NPRs upon signing a partnership agreement with the GoN, a further 150,000 NPRs after completing the foundation, and a final 100,000 NPRs after completing the walls. A retrofitting grant of 100,000 NPRs is also available to households whose homes have been partially damaged. This grant is provided in two tranches of 50,000 NPRs. The National Reconstruction Authority (NRA) (National Reconstruction Authority) and HRRP defined a core package of socio-technical assistance activities to complement the GoN financial assistance.

By the middle of September 2018, the NRA reported that of the 810,196 households eligible for the 300,000 NPRs GoN housing reconstruction grant, 89% have now signed the partnership agreement with the GoN, 88% have received the first tranche (50,000 NPRs), 61% have received the second tranche (150,000 NPRs), and 32% have received the third tranche (100,000 NPRs). Of the 50,784 households eligible for the GoN housing retrofit grant, 18% have signed the partnership agreement with the GoN, 16% have received the first tranche (50,000 NPRs), and just one household has received the second tranche (50,000 NPRs). Through these grants, the GoN has so far disbursed more than 1.35 billion USD to earthquake affected households.

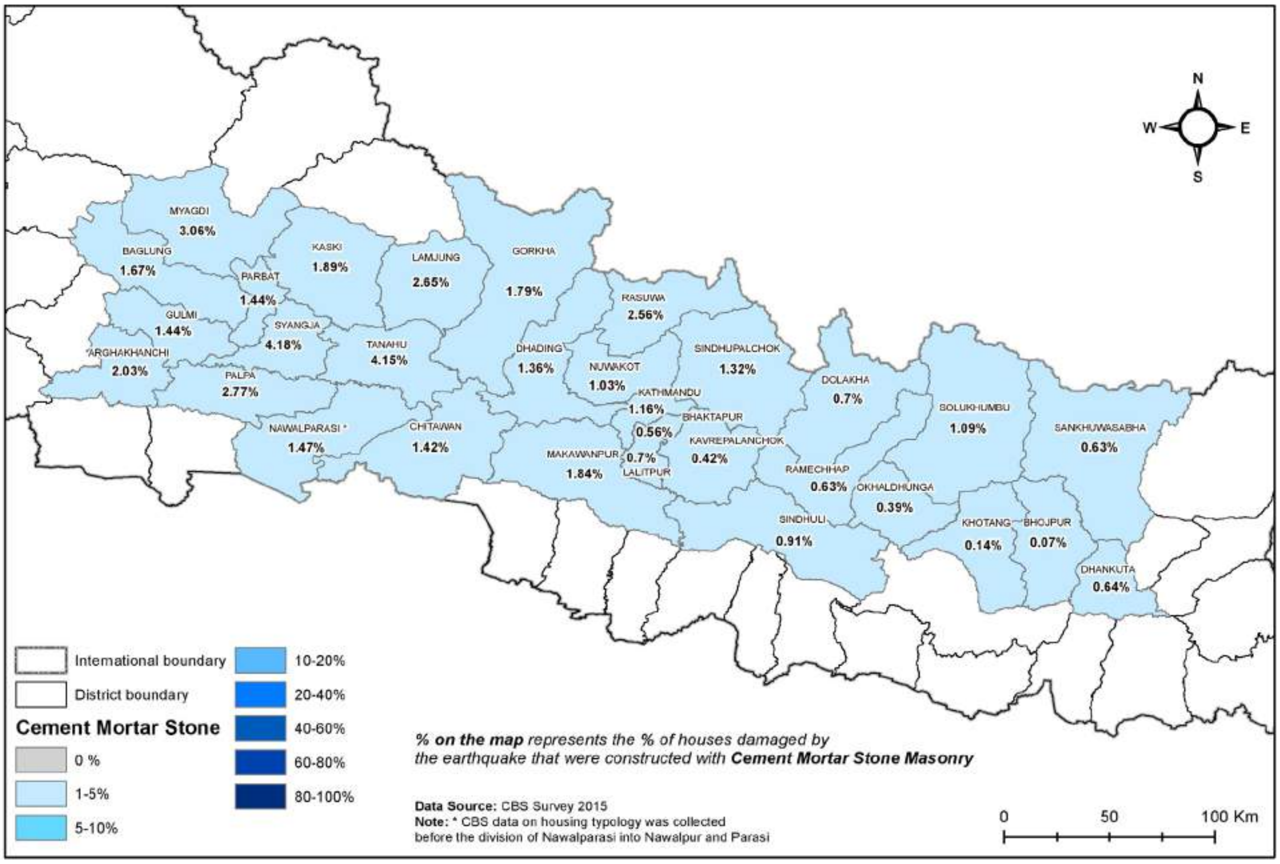
Where the information is available, the tranche status is provided in the photo captions.

The GoN has deployed more than 3,000 technical staff (engineers, sub-engineers, and assistant sub-engineers) across the 32 earthquake affected districts to manage in the inspection process associated with tranche disbursement. Where relevant, these technical staff are referred to in the photo captions as 'government engineers'.

1.0 Stone and Cement Mortar Masonry

In these types of buildings, the wall thickness can be reduced compared to stone and mud mortar masonry and is normally 350-450mm. The walls are load bearing. The building code allows for up to 2.5 storeys in stone and cement mortar masonry. In urban areas the floors and roofs are constructed with either reinforced concrete or timber. In rural settings the floors and roof structures are usually made with timber. In some cases, a sloping roof constructed from Reinforced Concrete (RC) slabs may be used. Usually the slabs rest directly on walls without beams. A peripheral beam at plinth and roof level can be found in some buildings. The percentage of buildings damaged in the earthquake that were stone and cement mortar masonry was quite low.

There was relatively limited prevalence of this typology pre-earthquake (see the map below).



Proportion of earthquake affected houses that were stone and cement mortar masonry pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• Use of local material (stone) • Where cement mortar is used, stone houses can be constructed up to 2.5 storeys (for stone and mud mortar masonry the limit is one storey) which is more appropriate for people's living requirements • High compressive strength • The mix of mortar may not be as per specifications and may not be standard throughout	• Cement may be difficult to access / transport / storage and may be expensive • Masons / construction workers may not be experienced in working with cement mortar • The mix of mortar may not be as per specifications and may not be standard throughout

Location: *Gorkha, Dharche Rural Municipality, Ward No. 2*

Tranche Status: *recommended for second tranche.*

Location: *Dhading, Netrawati Rural Municipality, Ward No. 5*

Tranche Status: *awaiting third tranche.*

Other: *the cost of constructing the house was approximately 600,000 NPRs.*

Location: *Kavrepalanchok, ChauriDeurali Rural Municipality, Ward No. 5*

Tranche Status: *unknown*

Other: *the family chose to build in stone and cement mortar masonry so that they could build sufficient size and storeys to meet their needs.*

Location: *Makwanpur, Manhari Rural Municipality, Ward No. 5*

Tranche Status: *approved for third tranche.*

Location: *Okhaldhunga, Molung Rural Municipality, Ward No. 5*

Tranche Status: *received third tranche*

Other: *two room, one storey house*

Location: *Ramechhap, Umakunda Rural Municipality, Ward No. 3*

Other: *a demonstration house constructed by the Gurkha Welfare Society (GWS).*

Location: *Rasuwa, Amma Chhodingmo Rural Municipality, Ward No. 5*

Tranche Status: *approved for third tranche.*

Other: *one room, one storey house which appears to be built adjacent to a larger dry stone masonry house.*

Location: *Sindhuli, Phikkal Rural Municipality, Ward No. 3*

Tranche Status: *approved for third tranche.*

Other: *two room, one-storey structure.*

Location: *Lalitpur, Dalchoki Municipality, Ward No. 3*

Tranche Status: *unknown*

Other: *salvaged material was used for the construction of this house.*

Location: *Lamjung, Sundar Bazar Municipality, Ward No. 4*

Tranche Status: *financial support provided by Choice Humanitarian Nepal*

Other: *two room, stone and cement mortar masonry house with RCC banding. This house was built as per one of 17 designs from the Department of Urban Development and Building Construction (DUDBC) design catalogue volume 1.*

Location: *Myagdi, Annapurna Rural Municipality*

Tranche Status: *approved for second tranche.*

Location: *Chitwan, Khairahani Municipality, Ward No. 2, Ladari*

Tranche Status: *unknown*

Location: *Parbat, Modi Rural Municipality, Ward No. 5*

Tranche Status: *first tranche received.*

Other: *this 1.5 storey house is non-compliant and the house owner has been advised to carry out correction measures to receive the remaining tranches. The government engineers recommended that timber vertical posts, timber banding at foundation and sill level, and RC banding at gable band be added.*

Location: *Syangja, Bhirkot Municipality, Ward No. 5*

Tranche Status: *first tranche received*

Other: *non-compliant two storey house requiring corrections to get the subsequent tranches. The house owner is not interested to carry out the corrections as they have finished their construction and they feel it is the government's fault for not providing technical assistance during construction period.*

Location: *Rasuwa, Uttargaya Rural Municipality, Ward no. 4*

Tranche Status: *received all three tranches*

Other: *one room, one storey, load bearing stone and cement mortar masonry house. Financial support for the grant was provided by Lumanti Support Group for Shelter.*

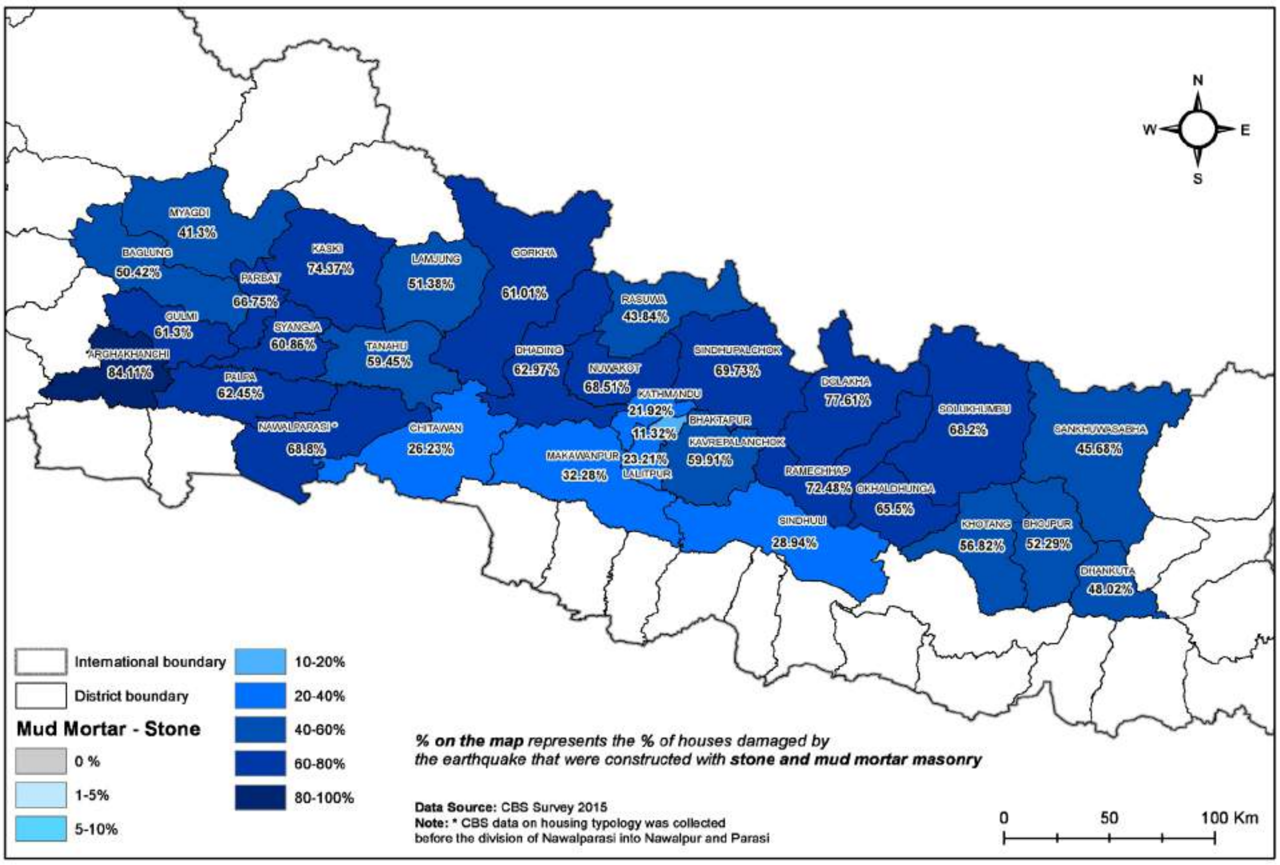
Photo credit: *Lumanti.*

Location: *Nuwakot, Tadi Rural Municipality, Ward No. 3*

Tranche Status: *received all three tranches*

2.0 Stone and Mud Mortar Masonry

This is the most common housing typology in the earthquake affected districts. The walls are load bearing and are usually 450-650 mm thick. These thick walls are usually constructed in two skins or layers (wythes), with smaller pieces of stone used as filler material between them. The flooring is normally bamboo or timber joists with mud plaster or flooring. Roofs are normally constructed using timber rafters with CGI sheets or tiles. By far, the largest proportion of buildings damaged in the earthquake were stone and mud mortar masonry.



Proportion of earthquake affected houses that were stone and mud mortar masonry pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• Use of local materials • Traditional way of building • Preserves vernacular architecture • Possible to salvage and reuse materials	• Limited under building code to a maximum of 1.5 storeys which does not fit with people's living requirements • Difficult to provide required bond • Cannot be used for walls of less than 12" thickness

Location: *Gorkha, Barpak, Sulikot Rural Municipality, Ward No. 3*

Tranche Status: *approved for second tranche*

Other: *RC banding used.*

Location: *Gorkha, Chumnubri Rural Municipality, Ward No. 4*

Tranche Status: *received second tranche*

Other: *one room, one storey houses with timber bands.*

Location: *Gorkha, Chumnubri Rural Municipality, Ward No. 4*

Tranche Status: *received all three tranches*

Other: *one room, one storey house in a very remote area where it is difficult to conduct multiple inspection visits. There are some technical issues with the house such as the stone gables and wide openings that look to be potentially non-compliant.*

Location: *Kavrepalanchok, Mahabharat Rural Municipality, Ward No. 3*

Tranche Status: *unknown*

Other: *one room, 1.5 storey house built to access grant but family still live in partially damaged pre-earthquake house and plan to continue living there.*

Location: *Makwanpur, Makwanpurgadhi Rural Municipality, Ward No. 8*

Tranche Status: *unknown*

Location: *Makwanpur, Makwanpurgadhi Rural Municipality, Ward No. 8*

Tranche Status: *received all three tranches*

Other: *two room, one storey house that cost 600,000 NPRs to build*

Location: *Nuwakot, Kispang Rural Municipality, Ward No. 1*

Tranche Status: *received second tranche*

Location: *Okhaldhunga, Chishankhu Gadhi Rural Municipality, Ward No. 3*

Tranche Status: *received all three tranches*

Other: *this was the first house reconstructed in this ward.*

Location: *Ramechhap, Likhu Tamakoshi Rural Municipality, Ward No. 2*

Tranche Status: *received second tranche*

Other: *timber banding is being used. Kam For SUD Nepal is providing socio-technical assistance.*

Location: *Rasuwa, Naukunda Rural Municipality, Ward No. 5*

Tranche Status: *approved for third tranche*

Other: *one room, 1.5 storey house with RC bands*

Location: *Sindhuli, Sunkoshi Rural Municipality, Ward No. 5*

Tranche Status: *received the second tranche but has not been approved for the third tranche.*

Other: *the house was not approved for the third tranche because the attic height exceeds specifications and the horizontal bands are not placed at the required spacing.*

Location: *Lalitpur, Bagmati Rural Municipality, Ward No. 4, Gimdi*

Tranche Status: *unknown*

Location: *Baglung, Jayamuni Municipality, Ward No. 3*

Tranche Status: *first tranche received*

Other: *non-compliant house constructed without technical assistance. Corrections need to be applied for the house owner to receive the remaining tranches. The government engineers are not able to provide sufficient guidance to the house owner on applying corrections as they have not received training for this.*

Location: *Syangja, Waling Municipality, Ward No. 5*

Tranche Status: *N/A*

Other: *this stone and mud mortar masonry house has survived three earthquakes; one in 1934, one in 2011, and the April 2015 Gorkha earthquake. The house is still standing despite the earthquakes and being more than 80 years old. Timber bands are provided at floor and roof level i.e. 2 bands per floor.*

Location: *Solukhumbu, Solu Dudhkunda Municipality, Ward No. 11, Tingla*

Tranche Status: *received first tranche*

Other: *non-compliant two storey house. The house owner needs to apply corrections to be approved for the second and third tranches. The house owner does not want to carry out the corrections as they feel the government is at fault for not providing technical assistance during the construction.*

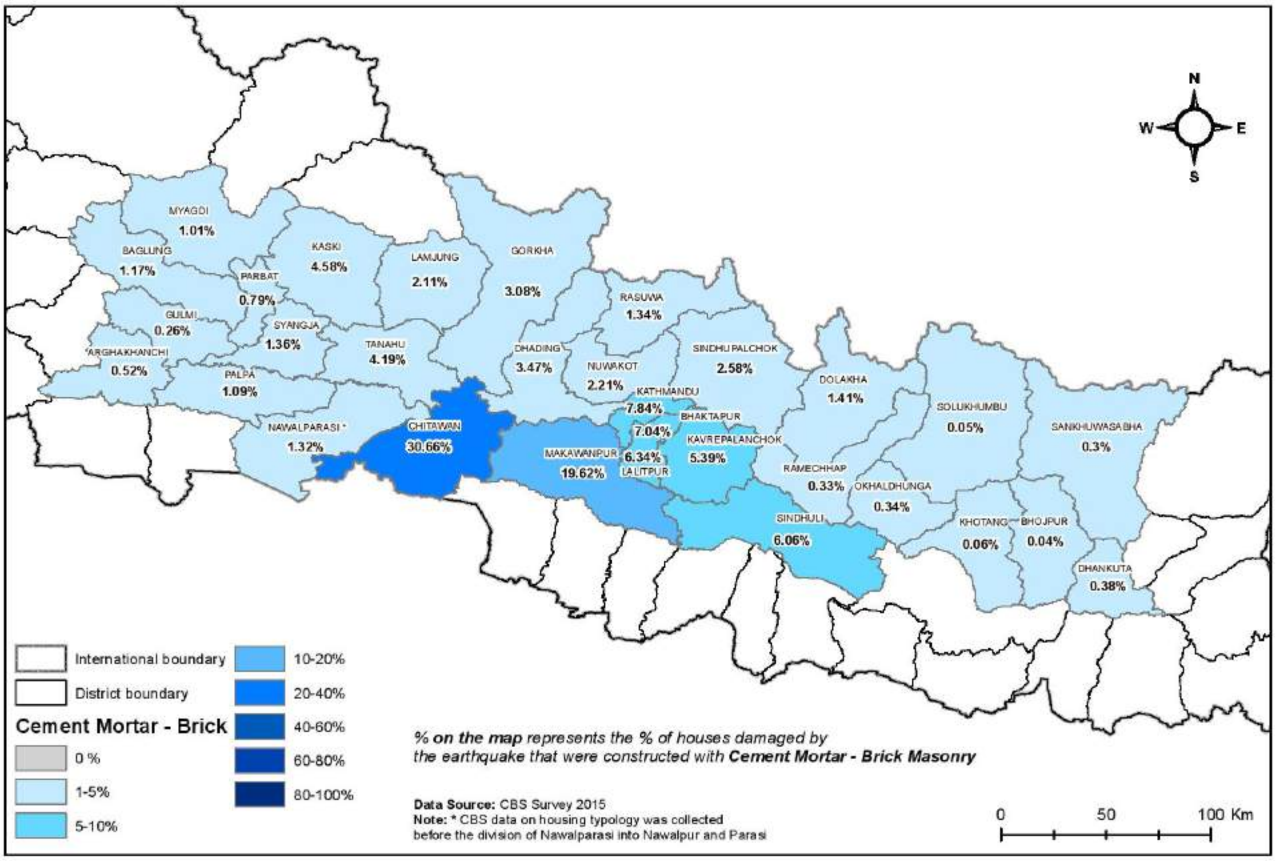
Location: *Parbat, Modi Rural Municipality, Ward No. 6*

Tranche Status: *unknown*

Other: *a one storey house with 3 rooms. Corrections need to be applied as the two parts of the structure need to be separated with an expansion joint and a buttress wall needs to be provided in the largest room as the dimensions exceed allowed limits.*

3.0 Brick and Cement Mortar Masonry

The walls are load bearing in these buildings. Construction is double brick with no cavity, around 230mm thick. These buildings are normally two or three storeys but can be even higher in some areas. The floor structure is usually timber and sometimes reinforced concrete (generally in urban areas). The roof is built with similar materials but in some cases a sloped roof is built using RC slabs. When slabs are used, they rest directly on walls without beams. A peripheral beam cast with the floor slab can be found in some buildings. The percentage of buildings damaged in the earthquake that were brick and cement mortar masonry was quite low.



Proportion of earthquake affected houses that were brick and cement mortar masonry pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• Easy to provide the required bond • Bonding strength is very good	• Bricks may be expensive due to transportation costs • Ground floor walls must be at least 230mm thick, so a large amount of bricks are required • The quality of the mortar may not be well controlled and can have a negative impact on the building strength

Location: *Makwanpur, Bhimfedi Rural Municipality, Ward No. 3*

Tranche Status: *unknown*

Location: *Makwanpur, Raksirang Rural Municipality, Ward No. 3*

Tranche Status: *received all three tranches*

Location: *Okhaldhunga, Maney Bhanjang Rural Municipality, Ward No. 7*

Tranche Status: *received all three tranches*

Other: *RC bands have been used.*

Location: *Kathmandu, Shankarapur Municipality, Ward No. 1*

Tranche Status: *not approved for second tranche*

Other: *the house owner does not have a building permit or house design. The house was not approved for the second tranche because the room size exceeds allowed limits and corrections are required.*

Location: *Sindhupalchok, Paanchpokhari Thangpal Rural Municipality, Ward No.9, Thangpaldhap, Taar*

Tranche Status: *unknown*

Other: *the house owner spent approximately 1,000,000 NPRs on the construction.*

Location: *Sindhupalchok, Nawalpur Rural Municipality, Ward No. 6*

Tranche Status: *received second tranche*

Other: *RC bands have been used.*

Location: *Dolakha, Jiri Municipality, Ward No. 6*

Tranche Status: *received first tranche*

Location: *Okhaldhunga, Siddhicharan Municipality, Ward No. 4*

Tranche Status: *received all three tranches*

Location: *Baglung, Baglung Municipality, Ward No. 3*

Tranche Status: *unknown*

Other: *the house owner is taking support from the government engineers for the construction of their house.*

Location: *Lamjung, Rainas Municipality, Ward No. 5*

Tranche Status: *unknown*

Other: *this house is not fully compliant as vertical reinforcement at the openings is missing.*

Location: *Parasi, Sunawal Municipality, Ward No. 2, Mukhiya Tole*

Tranche Status: *received second tranche*

Other: *vertical and horizontal reinforcement has not been properly included so the house has not been approved for the third tranche. The government engineer has recommended corrections, but the house owner does not want to carry out the corrections.*

Location: *Parbat, Kusma Municipality, Ward No. 4*

Tranche Status: *unknown*

Other: *the house owner, a single woman, has constructed a one room house with brick and cement mortar masonry (on the right of the photo) and is planning to demolish her old house (in the background) since it is heavily damaged. For now, she is still living in the old house.*

Location: *Tanahun, Bhanu Municipality, Ward No. 2*

Tranche Status: *received first tranche*

Other: *home owner does not know anything about the process to receive the second tranche of the grant. The house has not yet been inspected by the government engineers.*

Location: *Parbat, Modi Rural Municipality, Ward No. 6*

Tranche Status: *unknown, eligible for retrofit grant*

Other: *the house owner has repaired the house but has not been able to retrofit it as the engineers in the district have not received any training on retrofitting and are unable to provide solutions.*

Location: *Parbat, Modi Rural Municipality, Ward No. 6*

Tranche Status: *received first tranche*

Other: *house is non-compliant and house owner needs to apply corrections to receive the second and third tranches.*

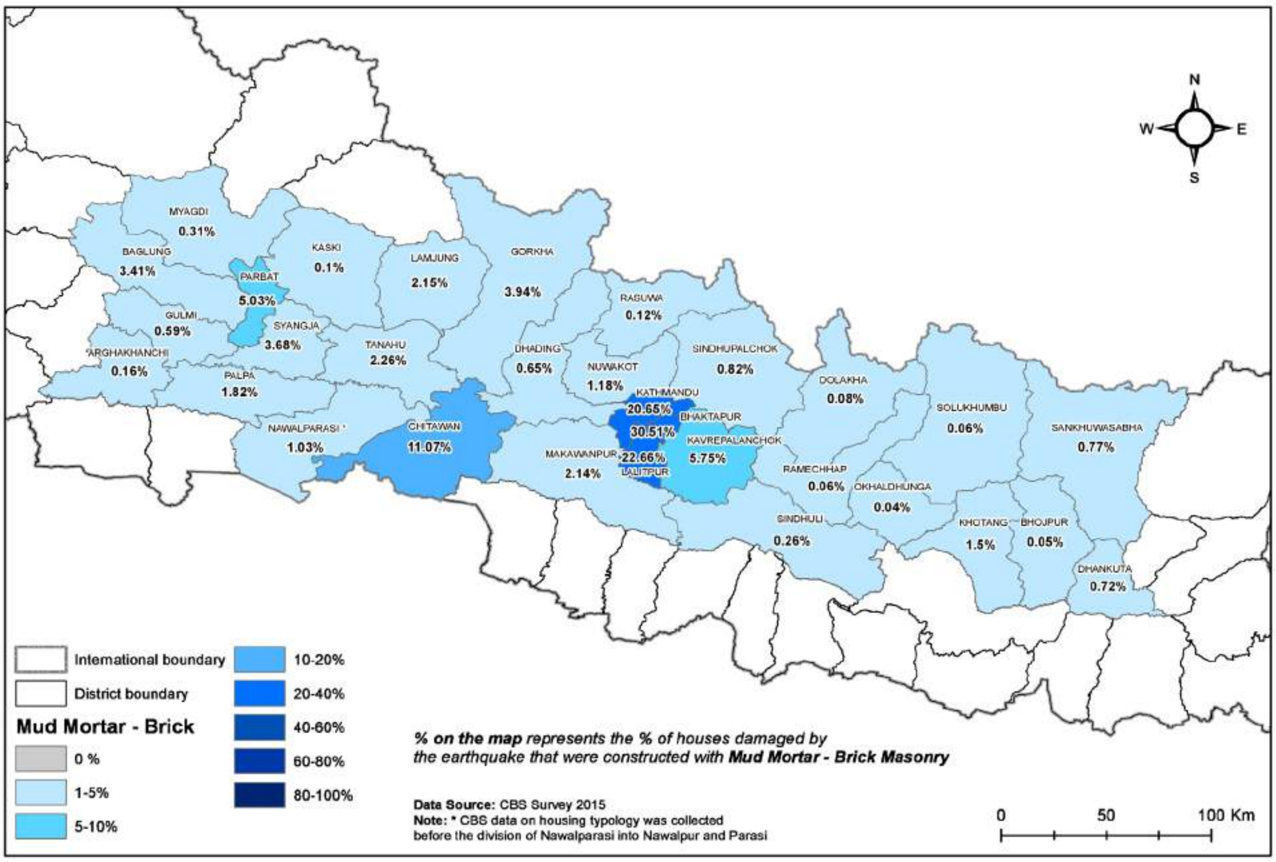
Location: *Makwanpur, IndraSarowar Rural Municipality, Ward No. 6*

Tranche Status: *approved for third tranche*

4.0 Brick and Mud Mortar Masonry

The structural walls are the main load carrying component of these buildings. Fired bricks are used as the building unit with mud mortar used as binder. These buildings are limited by the building code to one storey plus an attic. The walls are usually 350-450 mm thick. The floor system is usually mud on bamboo or timber joists and beams and roof structures are timber or steel with CGI, tile or slate cladding.

The percentage of buildings damaged in the earthquake that were brick and mud mortar masonry was low compared to other typologies, but there were much higher proportions of damage to these buildings in the Kathmandu Valley. This is due to brick and mud mortar masonry being the traditional style of construction in the old core areas of the valley, which were heavily affected by the earthquake. It is also worth noting that traditionally these buildings are three or more storeys which exceeds the building code limit of one storey plus attic.



Proportion of earthquake affected houses that were brick and mud mortar masonry pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• Possible to salvage and reuse materials • Traditional typology in some areas	• As the walls are usually 350-450mm thick, a lot of bricks are required • Low tensile strength • The building code limits construction to one storey plus attic which does not fit with people's living requirements

Location: *Nuwakot, Belkotgadhi Municipality, Ward No. 13*

Tranche Status: *received second tranche*

Other: *non-compliant and not approved for the third tranche as it exceeds the limit of one storey plus attic.*

Location: *Nuwakot, Suryagadhi Municipality, Ward No. 4*

Tranche Status: *received all three tranches*

Location: *Dhading, Salyantar*

Tranche Status: *unknown*

Other: *earthquake resilience elements such as bands are missing and the structure exceeds the building code limit of one storey plus attic.*

Location: *Bhaktapur, Madhyapur Thimi Municipality*

Tranche Status: *received second tranche*

Other: *two houses attached to each other belonging to brothers from the same family. Salvaged materials have been used.*

Location: *Lalitpur, Lalitpur Metropolitan City*

Tranche Status: *unknown*

Other: *traditional brick and mud mortar masonry house, built around a courtyard, near Patan Durbar Square. This house was built pre-earthquake and exceeds the building code limit of one storey plus attic (the building may predate the building code!).*

Location: *Lalitpur, Lalitpur Metropolitan City*

Tranche Status: *unknown*

Other: *traditional brick and mud mortar masonry house struggles for space between two newer buildings near Patan Durbar Square. This house was built pre-earthquake and exceeds the building code limit of one storey plus attic (the building may predate the building code!).*

Location: *Kathmandu, Kathmandu Metropolitan City*

Tranche Status: *unknown*

Other: *brick and mud mortar masonry buildings around the Kathmandu Durbar Square. These buildings were built pre-earthquake and exceed the building code limit of one storey plus attic (the buildings may predate the building code!).*

Location: *Bhaktapur, Madhyapur Thimi Municipality*

Tranche Status: *unknown*

Other: *traditional brick and mud mortar masonry buildings around a courtyard in Thimi. These buildings were built pre-earthquake and exceed the building code limit of one storey plus attic (the buildings may predate the building code!).*

5.0 Reinforced Cement Concrete (RCC) Frame

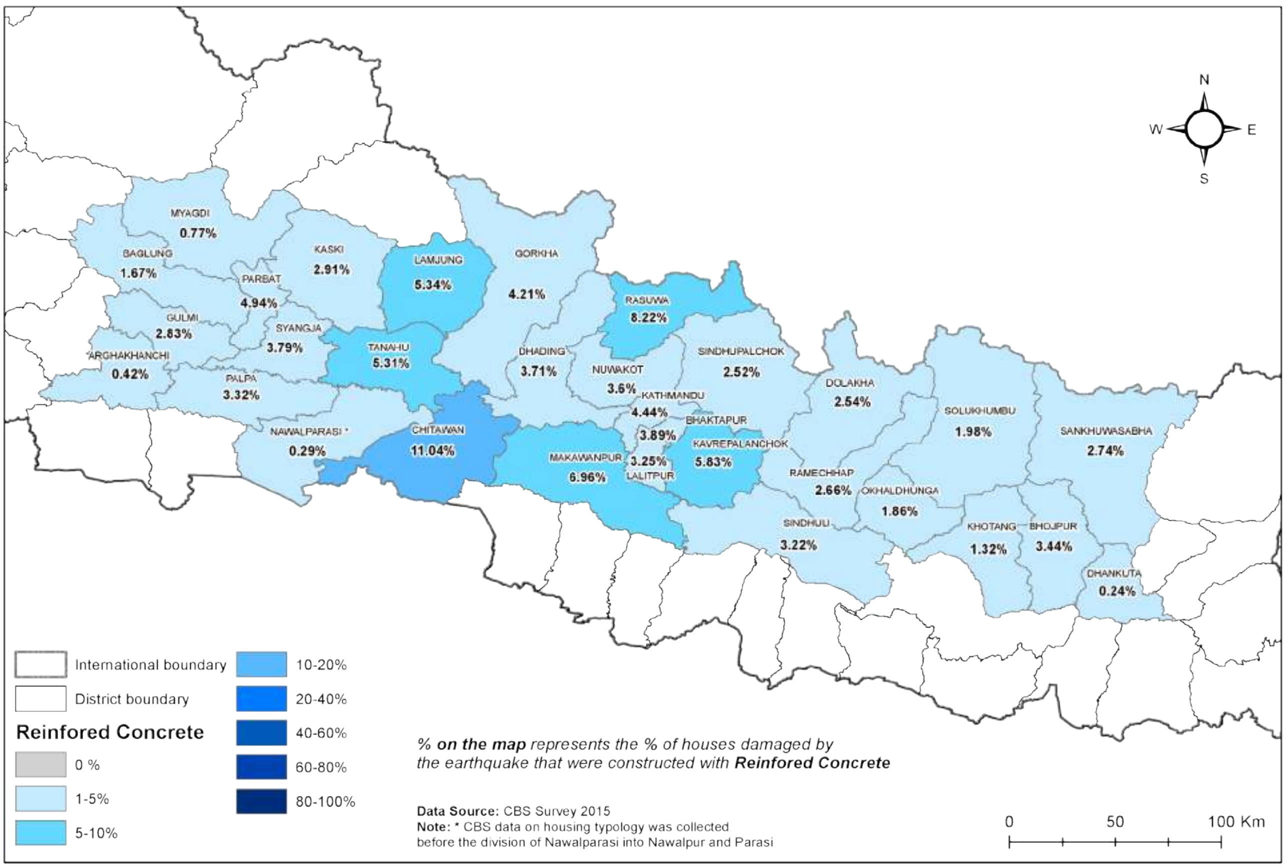
The frame of the RCC building is load bearing, however the walls play an important role in the structural integrity of the building. In general, the walls are brick or stone with cement mortar. There are two common ways to construct RCC buildings; the walls can be done first (engaged masonry), or the walls can come afterwards (brick infill).

Engaged masonry uses the walls as part of the structure, thereby reducing the required dimensions of the RCC frame. Infill masonry does not compliment the structure, and in fact serves to weaken the connections within the RCC frame, therefore requiring a much heavier RCC frame to compensate.

While RCC structures are most prevalent in urban and municipal settings, they are becoming more common in rural areas where access to engineering support and quality materials is limited. RCC structures not only perform best when engineered properly, but also require a higher level of craftsmanship than most brick or stone structures. The prevalence of non-engineered buildings is high in earthquake affected districts.

Floors and roofs are cast in RCC as part of the monolithic frame.

The figures and map below represent all types of RCC structures.



Proportion of earthquake affected houses that were RC pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• High compressive and adequate tensile strength • Fire and weather resistance • Durability	• Requires engineering input for design and construction • Takes a long time to build

Location: *Dhading, Thakre Rural Municipality*

Tranche Status: *unknown*

Other: *brick and cement mortar masonry infill used.*

Location: *Gorkha, Dharche Rural Municipality, Ward No. 2*

Tranche Status: *received first tranche*

Other: *stone and cement mortar masonry with RC bands used for the walls. The house owner has relocated from Kerauja to Lamabesi and has received the land ownership certificate (laal purja).*

Location: *Gorkha, Gorkha Municipality, Ward No. 8*

Tranche Status: *received second tranche*

Other: *brick and cement mortar masonry infill used for the walls.*

Location: *Kavrepalanchok, Panauti Municipality, Ward No. 11*

Tranche Status: *unknown*

Other: *the house owner is building this house with a skilled but untrained mason who has been working for more than 15 years. The other workers are all new masons. Some corrections had to be made to the columns before they were poured.*

Location: *Kavrepalanchok, Banepa Municipality, Ward No. 11*

Tranche Status: *received third tranche*

Other: *approved for third tranche following a non-compliance check carried out by the District Support Engineer (DSE) who did a full structural analysis of the house.*

Location: Makwanpur, Hetauda Sub-Metropolitan City, Ward No. 3

Tranche Status: *received first tranche*

Other: *brick and cement mortar masonry infill used for the walls.*

Location: *Makwanpur, Thaha Municipality, Ward No. 9*

Tranche Status: *received second tranche*

Other: *fully compliant house with 12"x12" columns. The reconstruction grant has been funded by Tearfund and socio-technical assistance has been provided by Lumanti.*

Location: *Okhaldhunga, Maney Bhanjang Rural Municipality, Ward No. 7*

Tranche Status: *received first tranche*

Other: *the house is non-compliant as it has been constructed with an eccentric footing and improper grid for the columns.*

Location: *Ramechhap, Likhu Tamakoshi Rural Municipality, Ward No. 2*

Tranche Status: *received first tranche*

Other: *a single bay RC frame house.*

Location: *Rasuwa, Gosaikunda Rural Municipality, Ward No. 4*

Tranche Status: *received first tranche*

Other: *a three storey, RC frame structure with stone masonry infill walls.*

Location: *Sindhuli, Sunkoshi Rural Municipality, Ward No. 6*

Tranche Status: *received all three tranches*

Location: *Dolakha, Baiteshwor Rural Municipality, Ward No. 6*

Tranche Status: *approved for second tranche*

Other: *a four room RC structure.*

Location: *Lamjung, Rainas Municipality, Ward No. 5*

Tranche Status: *unknown*

Other: *one bay RC frame structure with 12"X12" columns. The structure has two rooms and has been built to allow for the possible addition of extra storeys in future.*

Location: *Myagdi, Beni Municipality, Ward No. 1*

Tranche Status: *unknown*

Other: *the house owner took a loan from the bank for the construction at 25% interest per annum. They have taken support from the government engineer during construction.*

Location: *Chitwan, Khairahani Municipality, Ward No. 2, Ladari*

Tranche Status: *received all three tranches*

Other: *two storey, two room structure with 12"X12" columns, with potential for extension in future.*

Location: *Tanahun, Bhanu Municipality, Ward No. 2*

Tranche Status: *unknown*

Other: *due to road right of way conflict and formation of new municipality and the municipal building requirements the house owner is not allowed to construct a rigid floor, so they have used CGI for a temporary roof and have brick infill walls.*

Location: *Chitwan, Rapti Municipality, Ward No. 8*

Tranche Status: *first tranche received*

Other: *this is what is known locally as a 'one day' house. These houses are constructed using precast RC columns which are placed at the corners, connections, and partitions and then brick and cement mortar masonry is used as infill. The precast columns are installed 2-3 feet under the ground and there are no bands or vertical posts at openings. The houses get their name from the fact that the frame can be erected in just one day. The house has two storeys, with two rooms on each floor. It is reported that at least 30% of houses being reconstructed post-earthquake have used this design. As the houses are non-compliant, the house owners cannot progress beyond the first tranche of the GoN housing reconstruction grant.*

Location: *Solukhumbu, Solududhakunda Municipality, Ward No. 6*

Tranche Status: *recommended for third tranche*

Other: *the house has a septic tank which is a requirement of the municipality's building permit process.*

6.0 Hybrid Structures

These structures bring together a combination of building typologies or systems in one structure. While this can mean any combination of materials and structures, there is a prevalence of structures with a masonry first storey and subsequent storeys in steel or timber. It is these types of structure that we see in the images below.

The first storey masonry structure requires all the earthquake resistant elements such as horizontal and vertical bands. Connection details are often a challenge for these structures as it is combining two systems and two types of material.

Hybrid structures were not a category of building type documented through the CBS damage assessment (falling under 'other'), but hybrid structures are now found throughout all 32 earthquake affected districts. On 25 September 2017, the NRA published the Hybrid Structure manual to provide guidance on construction of these buildings in relation to the requirements under the reconstruction grant process.

Pros	Cons
• Lighter structure on upper storey(s) • Speed of construction	• Lack of clarity on compliance requirements for reconstruction grant • Potential for poor connections between different sections / materials which could weaken structure

Location: *Okhaldhunga, Molung Rural Municipality, Ward No. 2*

Tranche Status: *received second tranche*

Other: *the ground floor has been constructed using stone and mud mortar masonry and the first floor is timber frame with CGI sheeting. The house has not been approved for the third tranche as it was found to be non-compliant during inspection.*

Location: *Sindhuli, Hariharpurgadhi Rural Municipality, Ward No. 1*

Tranche Status: *approved for third tranche*

Other: *the ground floor has been constructed using stone masonry and the first floor is timber frame.*

Location: *Sindhuli, Hariharpurgadhi Rural Municipality, Ward No. 2*

Tranche Status: *approved for third tranche*

Other: *the ground floor has been constructed using brick masonry and the first floor is timber frame.*

Location: *Sindhuli, Tinpatan Rural Municipality, Ward No. 3*

Tranche Status: *received second tranche*

Other: *the ground floor has been constructed using brick masonry and the first floor is timber frame. The house has not been approved for the third tranche due to the timber first floor.*

Location: *Kavrepalanchok, Mahabharat Rural Municipality, Ward No. 8*

Tranche Status: *received second tranche*

Other: *constructed based on local knowledge and traditional building practices. The owner needs to apply some corrections to receive the third tranche.*

Location: *Kavrepalanchok, Khanikhola Rural Municipality, Ward No. 6*

Tranche Status: *unknown*

Other: *ground floor is constructed with stone and mud mortar masonry and the first floor will be timber frame. After the release of the NRA's Hybrid Structure Manual, the house owner opted for this design in consultation with the government engineers.*

Location: *Okhaldhunga, Siddhicharan Municipality, Ward No. 2, Nisankhe*

Tranche Status: *received second tranche*

Other: *hybrid structure with stone masonry ground floor and timber and metal sheeting used for the first floor.*

Location: *Kolki-Rainas Urban Municipality, Ward No. 5, Lamjung*

Tranche Status: *received second tranche*

Other: *ground floor constructed using stone and mud mortar masonry with RC bands, and first floor constructed using CGI and timber. This household was supported by Lutheran World Relief (LWR) / COPPADES up to second tranche, but LWR / COPPADES only had agreement to support up to 200,000 NPRs so the third tranche will come from the Government and there is a risk that the house will be non-compliant.*

Location: *Rasuwa, Gosainkunda Rural Municipality, Ward No. 4*

Tranche Status: *received first tranche*

Other: *a hybrid structure with dry stone masonry ground floor and timber frame first floor.*

Location: *Rasuwa, Parbati Kunda Municipality, Gatlang*

Tranche Status: *unknown*

Other: *these traditional structures are two storey and are constructed with three sides from dry stone masonry and the front face is timber frame.*

Location: *Nuwakot, Belkotgadi Municipality, Ward No. 13*

Tranche Status: *first tranche received*

Other: *the house has been found to be non-compliant. The ground floor has been built with stone and mud mortar masonry with RC bands and the first floor is brick and cement mortar masonry with RC bands.*

Location: *Dolakha, Jiri Municipality, Ward No. 6*

Tranche Status: *received second tranche*

Other: *brick and cement mortar masonry with RC bands used for ground floor and steel frame used for first floor.*

Location: *Kathmandu, Gokarneswor Rural Municipality, Ward No. 1*

Tranche Status: *unknown*

Other: *one storey, hybrid structure constructed from a mix of stone and cement mortar masonry and CGI sheeting.*

Location: *Myagdi, Annapurna Rural Municipality, Ward No. 6*

Tranche Status: *received first tranche*

Other: *this steel frame structure uses stone masonry for the infill walls and has a metal floor. The house owner completed construction of the house just 4 months after the 2015 earthquake. The government engineers have found the house to be non-compliant. The total cost of construction was approximately 1,000,000 NPRs and the house owner took a loan of 600,000 NPRs, at 25% annual interest for the construction. The house owner works as a construction and agriculture labourer earning 300 NPRs per day.*

Location: *Solukhumbu, Solu Dudhkunda Municipality, Ward No. 11, Tignasa*

Tranche Status: *received first tranche*

Other: *constructed with stone and mud mortar masonry on the ground floor and timber frame with CGI sheets for the first floor. RC bands have been provided at plinth level and timber bands have been used at the lintel and sill level. The opening sizes and locations are not appropriate.*

Location: *Solukhumbu, Solududhakunda Municipality, Ward No. 6*

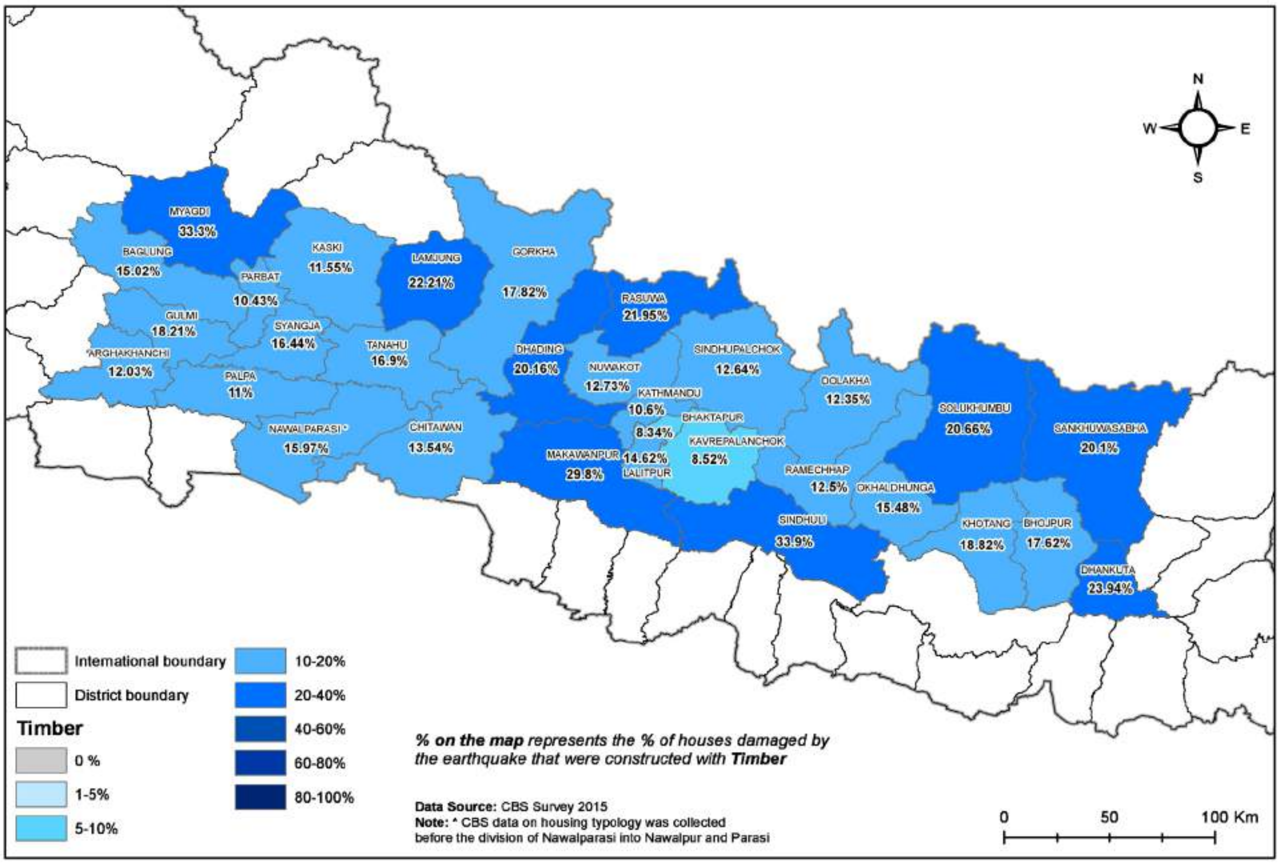
Tranche Status: *received first tranche*

Other: *non-compliant hybrid structure. The ground floor is RC frame with stone masonry infill, the first floor is RC frame with brick masonry infill, and the second floor is timber frame with metal sheeting and plywood walls. The floors are timber. The home owner is currently living in their earthquake damaged house behind the new house. This house will have to be demolished once the new one is complete due to the municipality road expansion.*

7.0 Timber Frame Structures

In these buildings, timber members are used as vertical load carrying units and the lateral load resistance system. Design considerations should be given to ensure correct timber species selection, minimal exposure of the timber elements to the weather and the timber members should be seasoned and treated before construction to minimize impacts of weather and insects. In timber frame buildings, the strength of the joints, the use of bracing, the moisture content and the species of timber has a significant impact on the overall strength of the structure.

In March 2018, the NRA published the 'Light Timber / Steel Frame Structure Manual' to provide guidance on construction of these buildings in relation to the requirements under the reconstruction grant process.



Proportion of earthquake affected houses that were timber frame pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• It's much quicker to build a timber frame house • Timber frame houses allow for a more flexible design • Timber frames can be built in sub-zero temperatures	• If not properly treated, timber frame can be susceptible to damp conditions • Potentially vulnerable to heat and fire

Location: *Kavrepalanchok, Mahabharat Rural Municipality, Ward No. 8*

Tranche Status: *unknown*

Other: *the house owner chose to construct a timber frame house as timber is readily available in the area. After the release of the NRA's Hybrid Structure Manual, the government engineers suggested that the home owner should use stone masonry as infill on the ground floor, so the house owner is now collecting stones for this.*

Location: *Kavrepalanchok, Khanikhola Rural Municipality, Ward No. 5*

Tranche Status: *unknown*

Other: *the house owner chose to rebuild with a timber frame structure as this is how the house was pre-earthquake.*

Location: *Rasuwa, Gosaikunda Rural Municipality, Ward No. 4*

Tranche Status: *received first tranche*

Location: *Sindhuli, Kamalamai Municipality*

Tranche Status: *unknown*

Other: *timber frame houses are particularly prevalent in Sindhuli. This structure is two storeys.*

Photo credit: *DFID Nepal*

Location: *Sindhuli, Kamalamai Municipality*

Tranche Status: *unknown*

Other: *timber frame houses are particularly prevalent in Sindhuli. This structure is two storeys.*

Photo credit: *DFID Nepal*

Location: *Dolakha, Jiri Municipality, Ward No. 5*

Tranche Status: *received second tranche*

Location: *Dolakha, Jiri Municipality, Ward No. 4*

Tranche Status: *received second tranche*

Location: *Dolakha, Kalinchowk Rural Municipality, Ward No. 1*

Tranche Status: *received all three tranches*

Other: *one storey, timber structure with light weight CGI sheeting as roof and wall covering. The CGI sheets have been used to cover the timber frame and protect the timber from rain.*

Location: *Dolakha, Kalinchowk Rural Municipality, Ward No. 1*

Tranche Status: *received first tranche*

Other: *two storey timber structure with dry stone foundation*

Location: *Sindhuli, Tinpatan Rural Municipality, Belghari, Ward No. 1*

Tranche Status: *unknown*

Other: *the foundation has been completed and the house owner is waiting for design drawings for the timber structure.*

Location: *Sindhuli, Hariharpurgadhi Rural Municipality, Ward No. 1*

Tranche Status: *received first tranche*

Other: *the foundation has been completed and the house owner is awaiting inspection for second tranche*

Other: *Timber treatment process*

Photo credit: *ASF Nepal*

Location: *Ramechhap*

Tranche Status: *unknown*

Other: *photo is from November 2015*

Photo credit: *GOAL*

Location: *Ramechhap*

Tranche Status: *unknown*

Other: *photo is from November 2015*

Photo credit: *GOAL*

Location: *Ramechhap*

Tranche Status: *unknown*

Other: *photo is from November 2015*

Photo credit: *GOAL*

Location: *Ramechhap, Shivalaya VDC*

Tranche Status: *unknown*

Other: *trekking lodges that have been rebuilt using timber and CGI, along the Everest Base Camp trekking route.*

8.0 Hollow Concrete Block Masonry

In these types of structures hollow concrete blocks are used as wall materials. Concrete blocks with a core void area larger than 25% of the gross area are considered hollow concrete blocks. Under the Nepal National Standards NS 119/2042, hollow concrete blocks are required to have a minimum compressive strength of 51 kg/cm2. HRRP conducted testing on 110 blocks from over 50 suppliers and 78% (86) were below the Nepal National Standards.

Hollow Concrete Block (HCB) masonry was not a category of building type documented through the CBS damage assessment (falling under 'other'), but HCBs are being used for reconstruction throughout all 32 earthquake affected districts. HCBs are particularly prevalent in Kaski, where production of HCBs is high.

A Technical Working Group (TWG) was launched in July 2018 to develop guidelines for construction with HCBs under the housing reconstruction grant process. The TWG includes representatives from NRA and Building CLPIU on the GoN side as well as representatives from the Partner Organisations (POs) Build Change, NRCS, and the National Society for Earthquake Technology-Nepal (NSET). The HRRP technical coordination team is facilitating the TWG.

Pros	Cons
• Less mortar required • Good thermal insulation properties	• Absence of proper construction guidelines • Poor quality of blocks being produced in districts • Lack of construction workers with experience working with HCBs

Location: *Kaski, Lekhnath Municipality, Pokhara*

Tranche Status: *unknown*

Other: *one of more than 600 houses that have been constructed using hollow concrete blocks in this area. House owners regularly ask the government engineers for designs for HCB houses. The engineers have been providing the designs from the DUDBC Design Catalogue Volume 2, but households are not building according to these.*

Location: *Parbat, Modi Rural Municipality, Ward No. 6*

Tranche Status: *first tranche received*

Other: *the house owner started construction of his 5 room house, with a steel frame structure and hollow concrete blocks for the infill walls, just 1 month after the 2015 earthquake. He is still going through the grievance mechanism and he didn't know that this house would not be in line with government standards as there was no guidance available previously and the inspection engineers were not deployed until December 2017.*

Location: *Dolakha*

Tranche Status: *unknown*

Other: *RC frame building with HCB infill.*

Location: *Nuwakot*

Tranche Status: *unknown*

Other: *HCB house with steel truss for roof.*

Location: *Rasuwa*

Tranche Status: *unknown*

Location: *Dolakha*

Other: *blocks are laid in the sun after production*

Location: *Dolakha*

Other: *blocks submerged in a pond for curing.*

Other: *various sizes and shapes of HCBs available*

Location: *Sindhupalchowk, Bhimtar*

Tranche Status: *N/A (constructed by World Vision)*

Other: *one storey, two room house constructed by World Vision.*

Location: *Makwanpur, Hetauda Sub-Metropolitan City, Ward No. 3*

Tranche Status: *unknown*

Location: *Gorkha*

Tranche Status: *unknown*

Location: *Nuwakot, Bidur Municipality, Battar, Ward No. 4*

Tranche Status: *received all three tranches*

Other: *house owner chose to construct with HCBs because they felt they were cheaper, more easily available, and provided better protection from the cold. The house owner had not received any technical assistance but the mason that worked on the construction had worked with HCBs previously.*

Location: *Parbat, Kusma Municipality, Ward No. 4*

Tranche Status: *received first tranche*

Other: *this house is non-compliant as it does not meet the requirements published in the two HCB type designs in the second volume of the DUDBC design catalogue. The government engineers are unable to provide solutions for corrections for non-compliant HCB houses.*

Location: *Lalitpur, Godawari Municipality, Goretar, Ward No. 10*

Tranche Status: *received first tranche*

Other: *this house was constructed by Shangri-La Orphanage Home (household should not have received first tranche in addition to house directly constructed by Shangri-La Orphanage Home).*

Location: *Lalitpur, Godawari Municipality, Lele, Tikabhairab, Ward No. 6*

Tranche Status: *unknown*

Other: *this house was supported by Batas Foundation. The house was found to be non-compliant during inspection as the gable band is missing and the sill and lintel bands are placed at a distance of just 3 feet.*

Location: *Lalitpur, Mahalaxmi Municipality, Lamatar, Sisneri, Ward No. 9*

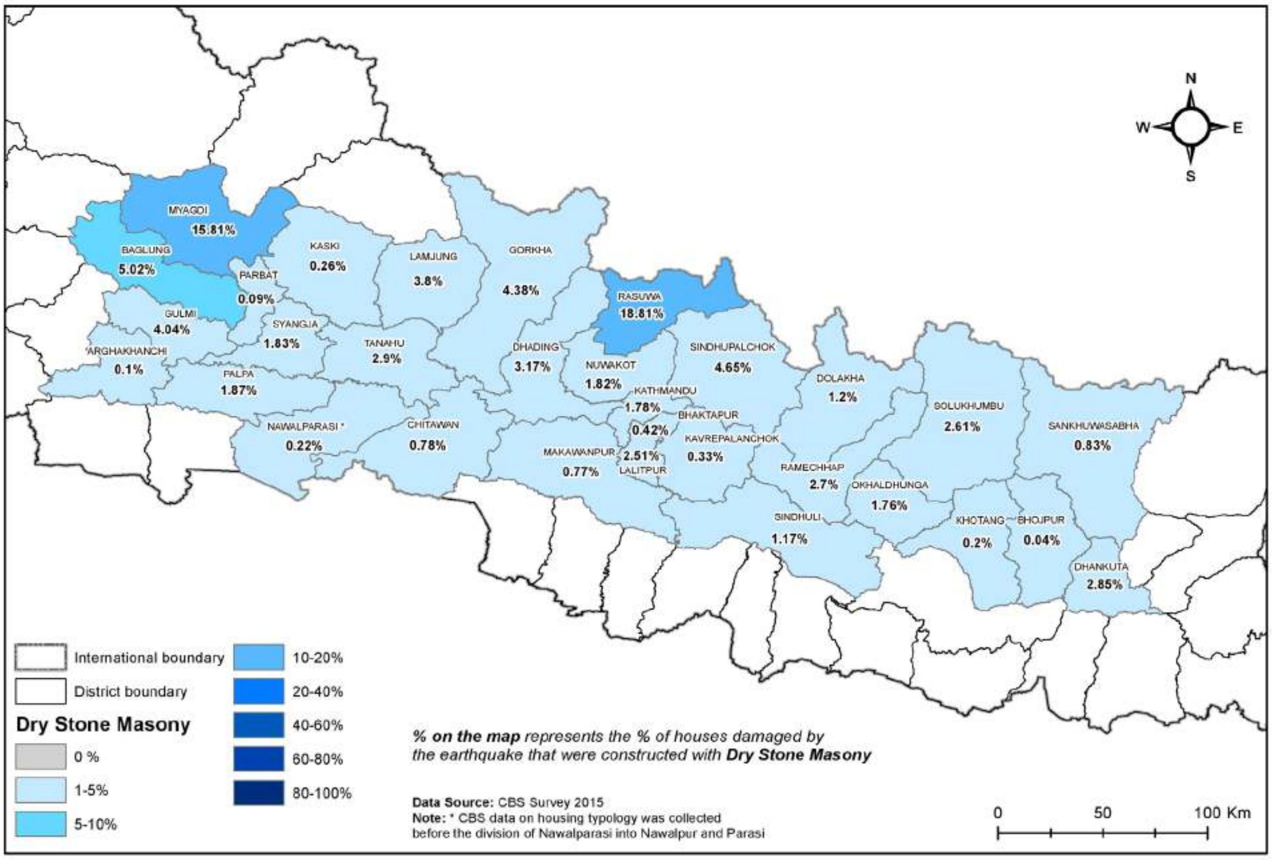
Tranche Status: *received first tranche*

Other: *the house owner got approval for the house design from the municipality. Approval for second tranche is pending.*

9.0 Dry Stone Masonry

This is a typology which is particularly common in the Northern parts of Nepal. In these structures, the walls are constructed using only stone and no mortar is used. For dry stone construction the stones are required to be better prepared or dressed as there is no mortar to accommodate gaps in the stones. There is evidence to suggest (reference NSET shake table tests) that dry stone construction performs better in an earthquake than mud mortar stone construction due to greater friction between stones which reduces the transference of ground forces up the wall. In some cases, mud is used after construction to fill the air gaps in the walls, but this is not structural.

In June 2017, the NRA published the inspection forms (first and second) for houses constructed using dry stone masonry so that these buildings could be brought into the housing reconstruction grant process.



Proportion of earthquake affected houses that were dry stone masonry pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• Use of local material • Traditional construction method • Suitable for climate in remote, northern areas (retains heat in winter)	• Large wall thickness • Absence of proper guidelines for construction

Location: *Dolakha, Kalinchowk Rural Municipality*

Tranche Status: *unknown*

Location: *Dolakha, Kalinchowk Rural Municipality, Ward No. 1*

Tranche Status: *received third tranche*

Location: *Gorkha, Kerauja, Dharche Rural Municipality, Ward No. 2*

Tranche Status: *first tranche recieved*

Other: *two storey house which has been found to be non-compliant. Correction measures are very difficult to apply so it is challenging for the house owner to progress through the reconstruction process. People construct two storey houses in this area as they need the space for their living requirements.*

Location: *Rasuwa, Parbati Kunda Municipality, Gatlang*

Tranche Status: *unknown*

Other: *reconstruction of houses in May 2015.*

Photo credit: *GOAL.*

Location: *Ramechhap, Bijulikot*

Tranche Status: *unknown*

Other: *Kumar Tamang and other masons working on the reconstruction of Tenjen Tamang's house.*

Photo credit: *Medair*

Location: *Dhading, Tipling*

Tranche Status: *unknown*

Other: *dry stone masonry house with no earthquake resistant elements.*

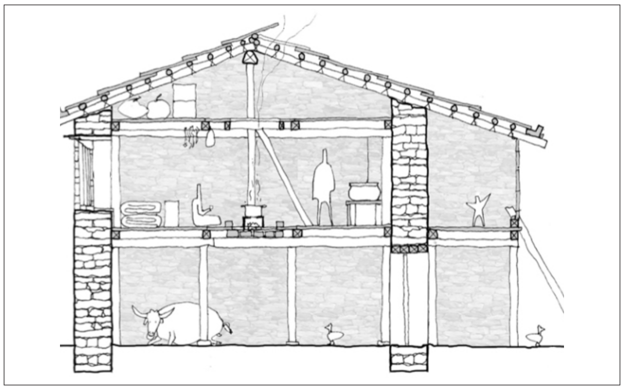
Photo credit: *NSET*

Location: *Gorkha, Lho*

Tranche Status: *unknown*

Other: *typical 'Khim' house.*

Photo credit: *CRS*



Other: *general elevation of typical 'Khim' house*

Credit: *CRS*

Location: *Dhading*

Tranche Status: *unknown*

Other: *traditional dry stone masonry houses with timber fronts. Traditional settlements in northern areas are generally compact and densely packed.*

Location: *Gorkha, Lho*

Tranche Status: *unknown*

Other: *traditional settlement pattern in Nothern areas.*

Location: *Dhading, Rubee Valley, Ward No. 1*

Tranche Status: *received second tranche*

Other: *one storey, one room house with all required earthquake resistant elements, including horizontal and vertical banding and double framed openings.*

Location: *Dhading, Rubee Valley, Ward No. 1*

Tranche Status: *received second tranche*

Other: *one storey, one room house with all required earthquake resistant elements, including horizontal and vertical banding and double framed openings.*

Location: *Dhading, Rubee Valley, Ward No. 1*

Tranche Status: *received first tranche*

Other: *one storey, one room house with all required earthquake resistant elements, including horizontal and vertical banding and double framed openings.*

Location: *Dhading, Rubee Valley, Ward No. 1*

Tranche Status: *received second tranche*

Other: *one storey, one room house with all required earthquake resistant elements, including horizontal and vertical banding and double framed openings.*

Location: *Dhading, Rubee Valley, Ward No. 1*

Tranche Status: *received first tranche*

Other: *one storey, one room house with all required earthquake resistant elements, including horizontal and vertical banding and double framed openings.*

Location: *Dhading, Rubee Valley, Ward No. 1*

Tranche Status: *received all three tranches*

Other: *one storey, one room house with all required earthquake resistant elements, including horizontal and vertical banding and double framed openings.*

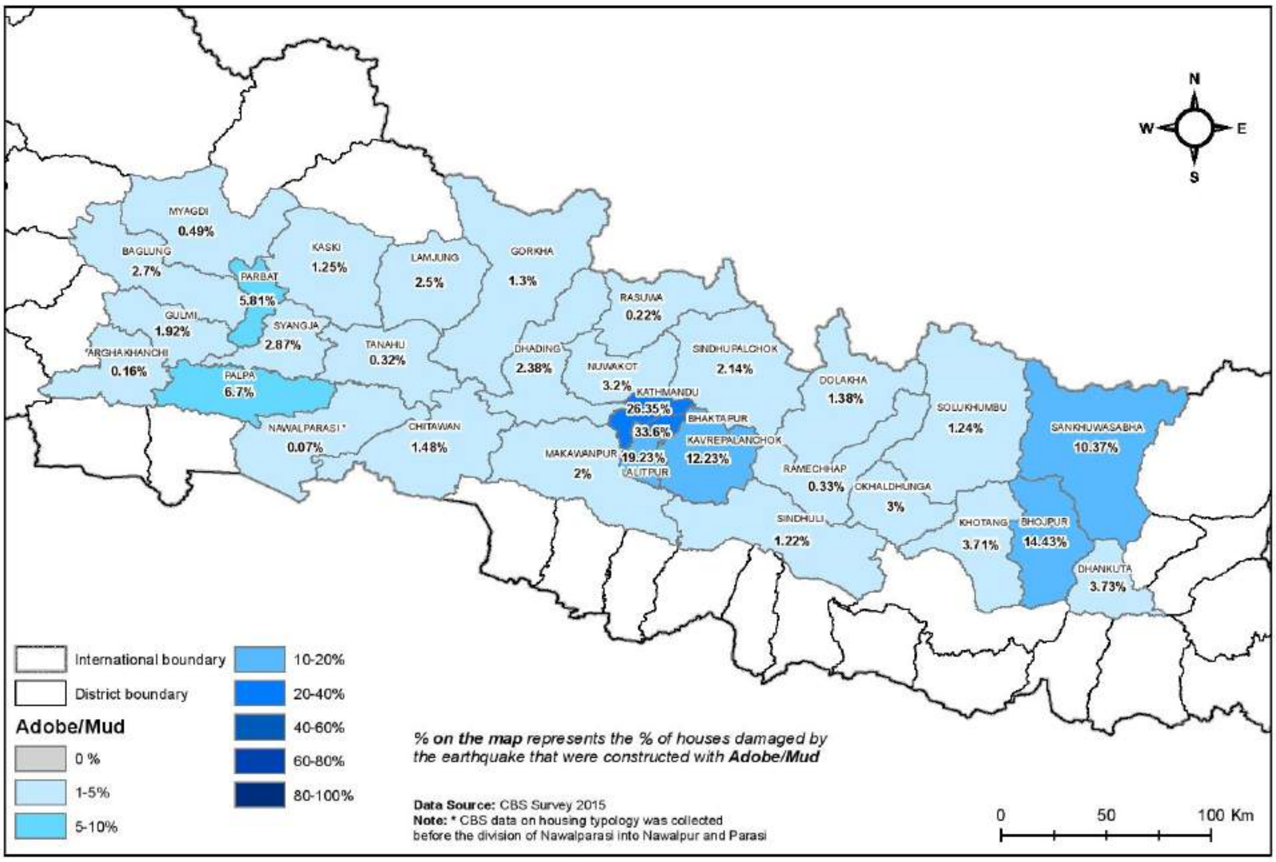
10.0 Adobe Structures

Adobe buildings are one of the oldest building typologies in Nepal. These are buildings constructed using earth. There are many varieties of adobe construction, ranging from earth blocks/bricks and cob to rammed earth systems. Some have load bearing walls and some use frames or other systems.

The most common form of adobe construction seen in earthquake affected districts is sundried earth bricks, usually with mud mortar. These have load bearing walls, but these bricks can also be used in other systems such as infill for framed structures, or a mix. Earthquake resistant elements such as banding and vertical ties (posts) are required for this type of structure.

Floors are often timber with mud overlay and roof structures are often timber with thatch, tile, or CGI cladding.

In November 2017, the NRA published the inspection form for houses constructed using sundried bricks so that these buildings could be brought into the housing reconstruction grant process.



Proportion of earthquake affected houses that were adobe/mud pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• Local Production of material • Good thermal performance	• Low strength • High level of maintenance required • Moisture reduces strength of materials over time

Location: *Kavrepalanchok, Panchkhal Municipality*

Tranche Status: *unknown*

Other: *this house was assessed by HRRP technical staff as part of the support to the NRA for the development of an inspection checklist for houses constructed with sun-dried adobe bricks. The house includes all required earthquake resistant elements and the house owner has also received a municipal permit for the design.*

Location: *Kavrepalanchok, Panchkhal Municipality, Ojhetar*

Tranche Status: *unknown*

Other: *a house constructed with "raw" (sun-dried, adobe) bricks with timber bands.*

Location: *Kavrepalanchok, Panchkhal Municipality*

Tranche Status: *unknown*

Other: *a house constructed with "raw" (sun-dried, adobe) bricks.*

Location: *Kavrepalanchok, Panchkhal Municipality*

Tranche Status: *unknown*

Other: *a house constructed with "raw" (sun-dried, adobe) bricks.*

Location: *Kavrepalanchok, Panchkhal Municipality*

Tranche Status: *unknown*

Other: *a house constructed with "raw" (sun-dried, adobe) bricks.*

Location: *Kavrepalanchok, Panchkhal Municipality*

Tranche Status: *unknown*

Other: *a house constructed with "raw" (sun-dried, adobe) bricks.*

Location: *Kavrepalanchok, Panchkhal Municipality*

Tranche Status: *unknown*

Other: *a house constructed with "raw" (sun-dried, adobe) bricks.*

Location: *Kavrepalanchok, Panchkhal Municipality*

Tranche Status: *unknown*

Other: *a house constructed with "raw" (sun-dried, adobe) bricks.*

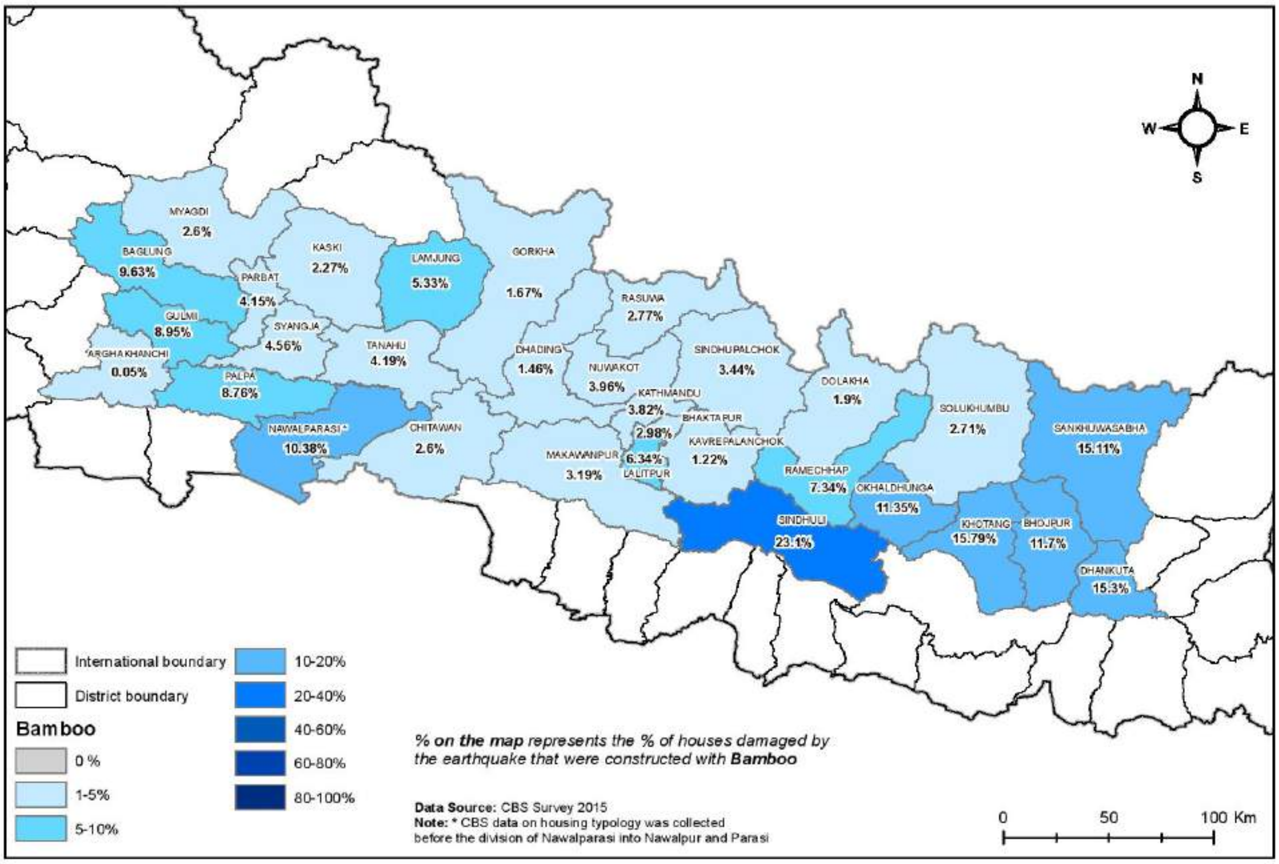
Location: *Tanahun, Bhanu Municipality*

Tranche Status: *first tranche received*

11.0 Bamboo

Bamboo is used in a range of typologies for floors and roofs, however there are also houses with bamboo walls, floors and roofs. These are most often framed structures with bamboo lattice cladding or other light weight material. They often have a short life span as treatment of bamboo is not common. Treated bamboo structures have a significantly longer life span. They are easy to build, extend and repair and are highly flexible in wind and earthquakes. However, the joining of bamboo members is a challenge and when done incorrectly can jeopardise the integrity of the structure.

There is currently no inspection checklist or manual for bamboo house construction within the government housing grant.



Proportion of earthquake affected houses that were bamboo pre-earthquake. Source: CBS Damage Assessment.

Pros	Cons
• Use of local materials	• Requires proper treatment to be structural
• Low cost	• Susceptible to pests if not treated
• Lightweight	

Location: *Jhapa (not an earthquake affected district)*

Tranche Status: *unknown*

Photo Credit: *Ana Gatoo/HFH Nepal*

Location: *Chitwan*

Tranche Status: *unknown*

Location: *Chitwan*

Tranche Status: *unknown*

Location: *Dhading, Arughat*

Tranche Status: *unknown*

Location: *Jhapa (not an earthquake affected district)*

Tranche Status: *unknown*

Photo Credit: *Ana Gatoo/HFH Nepal*

Location: *Jhapa (not an earthquake affected district)*

Tranche Status: *unknown*

Other: *mud plaster being applied to the structure*

Photo Credit: *Ana Gatoo/HFH Nepal*

Location: *Jhapa (not an earthquake affected district)*

Tranche Status: *unknown*

Other: *bamboo frame being erected*

Photo Credit: *Ana Gatoo/HFH Nepal*

Location: *Jhapa (not an earthquake affected district)*

Tranche Status: *unknown*

Other: *connection detail for bamboo frame*

Photo Credit: *Ana Gatoo/HFH Nepal*

12.0 Compressed Stabilised Earth Block (CSEB) Masonry

In these houses, Compressed Stabilised Earth Blocks (CSEBs) are used as the wall unit materials in the load bearing walls. CSEBs, commonly called 'pressed earth blocks', are produced by placing a mixture of inorganic subsoil, non-expansive clay, aggregates, and Portland cement under a high amount of pressure in a mould to form blocks.

Compressed Stabilised Earth Block (CSEB) masonry was not a category of building type documented through the CBS damage assessment. In general, there has been limited uptake of CSEB and it is primarily seen in areas where it has been promoted / provided by Partner Organisations (POs).

Pros	Cons
• Low environmental impact • Quick construction	• Challenges in quality control • Soil quality needs to be standard – not always possible

Location: *Lalitpur, Godavari Municipality, Ward No. 6*

Tranche Status: *received first tranche*

Other: *one storey, two room house directly constructed by Samaritan's Purse and their reconstruction partner Build Up Nepal (household should not have received first tranche in addition to house directly constructed by Samaritan's Purse).*

Location: *Ramechhap, Sunapati Rural Municipality, Hiledevi, Ward No. 3*

Tranche Status: *received first tranche, not approved for second tranche as they missed the July deadline for tranche disbursement*

Other: *two-room house under construction. The CSEB blocks are produced by a company that is providing free transportation of blocks within a 10km radius. The producer estimates that a two-room house can be constructed for 650,000 NPRs.*

Location: *Ramechhap, Sunapati Rural Municipality, Dimipokhari, Ward No. 2*

Tranche Status: *received second tranche*

Other: *two room house that cost 500,000 NPRs to construct.*

Location: *Sindhupalchok, Majhi Basti, Bhimtar, Indrawati Rural Municipality*

Tranche Status: *unknown*

Other: *house constructed by Sweta Shree Foundation.*

Location: *Nuwakot, Bhimsen Rural Municipality, Ward no.5*

Tranche Status: *N/A, house directly constructed by OXFAM*

Other: *constructed in an integrated settlement in Baguwa.*

Location: *Lalitpur, Godawari Municipality, Ward No. 6*

Tranche Status: *received first tranche*

Other: *two room house built by Samaritan's Purse with technical support from Build Up Nepal (household should not have received first tranche in addition to house directly constructed by Samaritan's Purse).*

Location: *Gorkha, Palungtar Municipality, Ward No. 8*

Tranche Status: *unknown*

Location: *Nuwakot, Bidur Municiaplity, Ward No. 6*

Tranche Status: *approved for third tranche*

Other: *supported by ICIMOD*

Location: *Nuwakot, Bidur Municiaplity, Ward No. 6*

Tranche Status: *approved for third tranche*

Other: *supported by ICIMOD*

Location: *Nuwakot, Bidur Municiaplity, Ward No. 6*

Tranche Status: *approved for third tranche*

Other: *supported by ICIMOD*

13.0 Light Steel Frame Structures

In these buildings, light steel hollow sections are used for vertical, horizontal, and diagonal framing to resist the loading. This frame is then covered with timber or CGI sheets. Usually these buildings are single storey and have CGI roofing.

Sometimes steel frame structures have masonry infill walls (which are not safe in earthquake zones), but this is covered under the hybrid structures section above.

In March 2018, the NRA published the 'Light Timber / Steel Frame Structure Manual' to provide guidance on construction of these buildings in relation to the requirements under the reconstruction grant process.

Pros	Cons
• Rapid construction • Lightweight structure	• Challenge to transport steel frame to site • Requires high level of technology and well-trained man power

Location: *Okhaldunga, Sidhhicharan Municipality*

Tranche Status: *unknown*

Photo Credit: *NRA/District Support Engineer*

Location: *Sindhuli, Kamalamai Muni Municipality*

Tranche Status: *unknown*

Photo Credit: *NRA/District Support Engineer*

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